Authenticated Name-Channel Noninterference in Frozen Language-Model Scoring

Adel Mansour | Research prototype and preprint draft | 3 November 2026

Abstract

Matched-name tests can detect changes in language-model hiring scores, but finite tests and learned suppression do not establish universal name invariance. We study a narrower information-flow property: changing explicitly designated identity text must not alter a deterministic public score when retained evidence is fixed. We harden an internal post-embedding gate with authenticated immutable review records, segmentation-independent scalarwise tokenization, a score-only interface, model-integrity checks and serialized inference. The resulting theorem is conditional on correct role labels and trusted runtime semantics; it does not require accurate CV grading. Two abstract symbolic gather specifications are unsatisfiable when asked for an identity-dependent retained coordinate, while two deliberately unsafe specifications yield countermodels. On one frozen affine-four-bit Qwen2.5-1.5B checkpoint, 108 synthetic name and segmentation interventions produced zero score differences and 3,255 directly equal public tensor comparisons. Five signed-request tampering variants and runtime drift controls were rejected. Scalarwise encoding increased evidence token counts by approximately 4.5-4.7 times on the three examples. This is a reproducible engineering artifact and preprint draft, not a universal semantic name-discovery system, a fairness certificate for race or a validated hiring tool.

1. Problem and precise contribution

The desired intervention changes only an applicant's name and leaves the documentary qualifications and job fixed. A score that changes under that intervention exhibits name sensitivity. That sensitivity is distinct from population racial discrimination: names are imperfect cues, and other retained features and policies can cause inequity. This work targets the name channel, not the whole sociotechnical hiring process.

There are three different claims. First, a finite experiment can find no name-caused decision changes. Second, a representation probe can fail to recover name identity. Third, an architecture can establish that designated identity inputs have no path to a public output. Neither of the first two alone proves the third. For example, the hidden vector H=(qualification, name) supports a head that reads only qualification; equal outputs do not erase the name from H.

Earlier local learned adapters reduced name-induced decision flips on synthetic CVs while internal name-ID probes remained successful. The present study does not reinterpret those empirical outcomes as a universal theorem. It introduces a separate structural information-flow mode and hardens its boundary. The contribution is an auditable system construction, failure-focused evaluation and explicit claim ledger, rather than a claim of a novel concept-erasure algorithm or literature-wide research novelty.

2. Related distinctions

Counterfactual fairness [1] motivates holding the individual's other modeled factors fixed while intervening on a sensitive attribute. Our typed-name intervention is narrower and does not instantiate a complete structural causal model of race. Information-flow systems separate confidentiality from integrity and explicitly identify a trusted computing base [2]. We apply that distinction to exclusion of identity activations and authentication of reviewed role labels.

LEACE [3] establishes a linear guardedness property while minimizing particular representation changes. It does not establish that this nonlinear program has forgotten every name or association. LoRA [4] and quantized-base adaptation [5] can alter internal computations with frozen base weights; finite adapter

training does not by itself prove all-name invariance. Concept bottlenecks [6] can restrict decision representations to supervised concepts, but a learned extractor can still be name-sensitive. Our current gate removes the designated identity channel before attention instead of inferring successful erasure from a finite concept/probe result.

3. Threat model

An untrusted requester controls serialized request bytes and may attempt to change reviewed roles, text, reviewer identifiers or policy versions; insert unusual names; fragment evidence fields; exceed limits; or overlap requests. A trusted reviewer/operator supplies role judgments, a reviewer-key map and a checkpoint manifest. The host process, Python/MLX implementation, deterministic kernels and original tokenizer are trusted. Malicious code already executing in the host can inspect buffers or keys and is not contained by Python naming conventions.

Every identity occurrence to be varied must have the identity role, including aliases, contacts and repeated references. Other text is evidence. A signature cannot prove those semantic labels correct. A valid reviewer can make an incorrect judgment; the tests preserve this negative control explicitly. Errors, timing, admission decisions, memory use, logs and private research observations are outside the public-score confidentiality claim. Thus the claim is accepted-input output noninterference, not termination-sensitive or constant-time secrecy [2].

4. Construction

4.1 Authenticated immutable review

A review payload contains the policy version, reviewer identifier, original source SHA-256 and ordered role/text nodes. HMAC-SHA256 authenticates its canonical JSON encoding, using at least 32 bytes of random secret key material. Verification uses constant-time digest comparison [7]. Duplicate JSON keys, extra schema fields, invalid Unicode and unsupported sizes are rejected. Accepted requests are immutable bytes; verified nodes are frozen tuples, so later caller mutation cannot alter the reviewed computation.

HMAC is shared-secret authentication, not asymmetric non-repudiation. The verifier holds the key and can issue reviews itself. The trusted review step must not be exposed as an applicant-accessible signing service. Experimental keys were randomly generated in memory and not saved; public CPU-test keys are fixtures and are not operational credentials.

4.2 Segmentation-independent encoding

The earlier fieldwise BPE codec did not preserve the same retained token sequence when evidence field boundaries changed. Four actual-tokenizer examples reproduce that limitation. The revised codec encodes each original Unicode scalar c independently with the original tokenizer t, defining C(s) as the concatenation of t(c) over s. It therefore satisfies $C(a+b)=C(a)\|C(b)$.

All original identity scalars are encoded as well. No anonymous replacement string is constructed. This scalarwise convention differs from ordinary native whole-string BPE and may affect model utility; its cost is measured rather than hidden under the phrase 'base weights unchanged.'

4.3 Internal exclusion and runtime

The model computes the original embeddings for all actual source tokens, including identity tokens. An internal gather retains only evidence coordinates in their original order. Only that compact tensor enters the transformer; causal masks and positions start from the compact sequence. No identity-derived cache, residual, padding or position survives into scoring. The original 28 transformer blocks, normalization and tied vocabulary projection remain unchanged. The scalar score is the A-versus-B logit margin and its constrained sigmoid.

The supported factory returns only an immutable two-field public score. Research tensor observations use a trusted backend subclass, not an untrusted audit flag. The runtime rejects external caches, checks the pinned architecture and parameter digest, and compares module/parameter-object identities before and after inference. A process-wide nonblocking lock serializes protected model calls across service objects. A backend failure disables the service until rebuilding; final closure rehashes model parameters and checkpoint files. Stable integer slots support name-independent tie breaking, conditional on name-independent slot assignment.

5. Conditional noninterference theorem

Let D be an accepted authenticated typed document and Q(D) its concatenated Unicode evidence text. Pointwise embedding lookup and the internal gather give:

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Z_0(D) = Emb(C(Q(D))).
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For any D and D' with Q(D)=Q(D'), the compact input tensors are equal, including their lengths. Each deterministic layer receives the same tensor, mask, positions and fixed weights. Induction yields Z_l(D)=Z_l(D') at every public layer; the fixed readout then yields PublicScore(D)=PublicScore(D'). No grading accuracy premise appears. The proof covers every accepted identity value and evidence resegmentation under the codec, not just the tested name list.

The HMAC adds a separate computational authenticity premise under its standard security assumptions. It does not strengthen the semantic completeness premise. Names left in evidence are outside the theorem. Conditionally, the downstream state is independent of the designated identity intervention given Q. Real population correlations between names and retained qualifications need not vanish, and other racial proxies remain possible.

The one-block and two-block SMT specifications use arbitrary arrays, arbitrary deterministic embedding coordinates, insertion positions and identity lengths. Z3 finds no differing retained coordinate (UNSAT). General finite-list extension and transformer induction are handwritten arguments. The solver does not verify Python, tokenizer behavior, floating-point kernels or role judgments end to end. An unsafe original-position specification admits a name-length countermodel; a finite-testing specification admits an untested-name failure (both SAT).

6. Experimental design

The frozen GPU plan was saved before inference and records source hashes, inputs, variants and acceptance checks. The design crosses three generated evidence bodies, 12 identity strings and three evidence/identity segmentation variants: 108 forwards. Strings include Arabic, Chinese, Devanagari and Korean, full-width letters, zero-width/NUL characters, an instruction-like string, a skill/name collision and a 1,000-character field. Names are synthetic treatment strings, not assigned race labels or real applicant records.

For each fixed evidence body, direct NumPy array comparisons check the compact embedding, all 28 block outputs, final normalization and full final-position vocabulary logits: 31 public tensor stages per forward. These comparisons do not substitute matching hashes for array equality. A separate check compares the manual block path with the original Qwen path on the same scalarwise evidence. Scalar scores must agree exactly across variants and remain nonconstant across evidence bodies.

Five request mutations test source, role, reviewer, policy and source-hash authentication. Actual runtime controls test overlay rejection, graph drift, serialization, failure poisoning and final integrity. Thirty CPU tests include 2,000 seeded random Unicode insertion/resegmentation cases. This is deterministic fuzzing, not an exhaustive Unicode proof or a representative sampling study.

The model is one pinned affine-four-bit Qwen2.5-1.5B-Instruct runtime on Apple Silicon MLX. It is distinct in precision from the original BF16 download. No adapter is trained or installed in this hardening study. Clef

and unrelated GPU training jobs are not used or modified. The exact model parameter digest, dependencies, run plan and code are in the artifact.

7. Results

All 108 forwards passed. Three forwards established per-body reference tensors; 105 counterfactual forwards yielded 3,255 directly equal public-stage comparisons within fixed-evidence groups and a maximum name/segmentation-induced score difference of zero. Different evidence bodies produced different margins: 0.1602668762, -0.0963649750 and -0.3481140137. This rejects a constant-only demonstration but does not establish grading correctness.

Measure	Result
Actual named-input forwards	108
Direct public tensor comparisons	3,255, all equal
Maximum matched score gap	0
Signed-payload tampering variants rejected	5/5
Real-tokenizer BPE segmentation failures reproduced	4
Revised codec segmentation checks	all equal
CPU tests	30 passed
Seeded Unicode property cases	2,000 passed
Safe SMT specifications	2 UNSAT
Unsafe SMT specifications	2 SAT

The runtime smoke test passed the final factory path, process-wide busy rejection, refusal to reuse a poisoned service and module-drift detection. Final parameter-content and checkpoint-file checks passed. No reviewer key was saved. Ordinary BPE required 59, 60 and 61 evidence tokens, compared with 271, 281 and 274 scalarwise tokens: 4.59x, 4.68x and 4.49x, respectively.

These counts are conformance evidence, not independent applicant trials. We do not attach a binomial population confidence interval to repeated forwards. The CPU result verifier reconstructs source hashes, factorial coverage and records; it does not independently re-run GPU inference or reconstruct omitted large tensors. Both this distinction and the abstract solver boundary are part of the reported result.

8. Limitations and publication status

The largest unresolved limitation is semantic identity-role correctness. A name may be an applicant, a client, a software identifier or a legitimate fact. A text-only program cannot guarantee two different correct roles for identical observable strings in different latent contexts. This does not prove that all learned invariant functions are impossible; it identifies an information gap the present system does not solve.

Signed incorrect reviews still permit identity leakage through evidence. Timing, codec caches, request lengths, rejection behavior and malicious host code are outside the output theorem. Graph/object checks and final hashes are not hardware attestation. Scalarwise encoding is token-expensive and its grading accuracy is unvalidated. External slot assignment or an application that breaks ties by name can invalidate an end-to-end ranking claim.

Only three synthetic evidence bodies and one model/runtime were studied. No independent reviewer study, external replication, real applicant-outcome study, multi-model replication or claim of new academic theory is supplied. The system is not ready for consequential hiring deployment. This manuscript is a preprint draft suitable for transparent portfolio presentation and further review, not a peer-reviewed publication or evidence that all racial bias has been eliminated.

9. Reproducibility and next experiments

The package contains source, frozen plans, solver specifications and outputs, complete synthetic intervention records, a read-only verifier, tests, a source/result manifest and a CPU CI recipe. Model weights and reviewer keys are excluded. The exact pin intentionally rejects other checkpoints. A new GPU run must use a new output copy rather than overwriting completed plans.

The next substantive studies should measure human role-annotation agreement and omitted-name error rates on separately consented CVs; replicate the structural conformance with independent code and another tokenizer/model; study more efficient segmentation-invariant codecs; and measure the utility, latency and admission costs. These are future work, not completed evidence.

Data, ethics and assistance disclosure

Only generated CV-like text is released in this extension. No race labels are inferred from individual names. The original project used public résumé/job datasets, but those records are not imported into this study or redistributed. Publishing this code is not a hiring deployment. Code, prose and analysis were developed with AI coding assistance; the artifact should be independently reviewed before academic submission. Model and dependency terms remain with their respective owners. The draft has no DOI or peer-review status.

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